

Unit 1.6 (Study guide) Section 1.3 (Stewart)

New functions from old functions.

- Transformations of functions.

Vertical and Horizontal Shifts

Suppose $c > 0$.

* $y = f(x) + c$, shift the graph
of $y = f(x)$ a distance c units
UPWARD

* $y = f(x) - C$, shift the graph
of $y = f(x)$ a distance C units
downward.

* $y = f(x - c)$, shift the graph
of $y = f(x)$ a distance c units to
the right

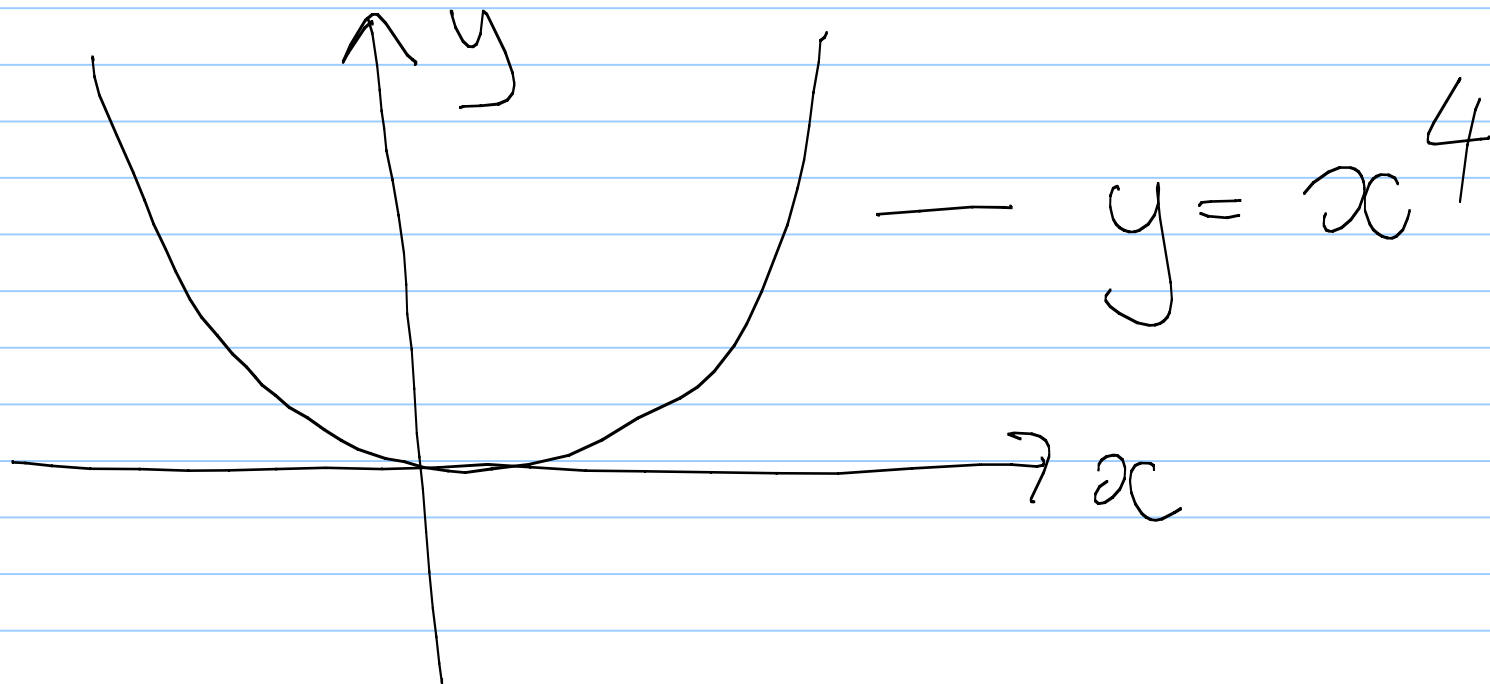
* $y = f(\underline{x+c})$, shift the graph of $y = f(x)$ a distance c units to the left.

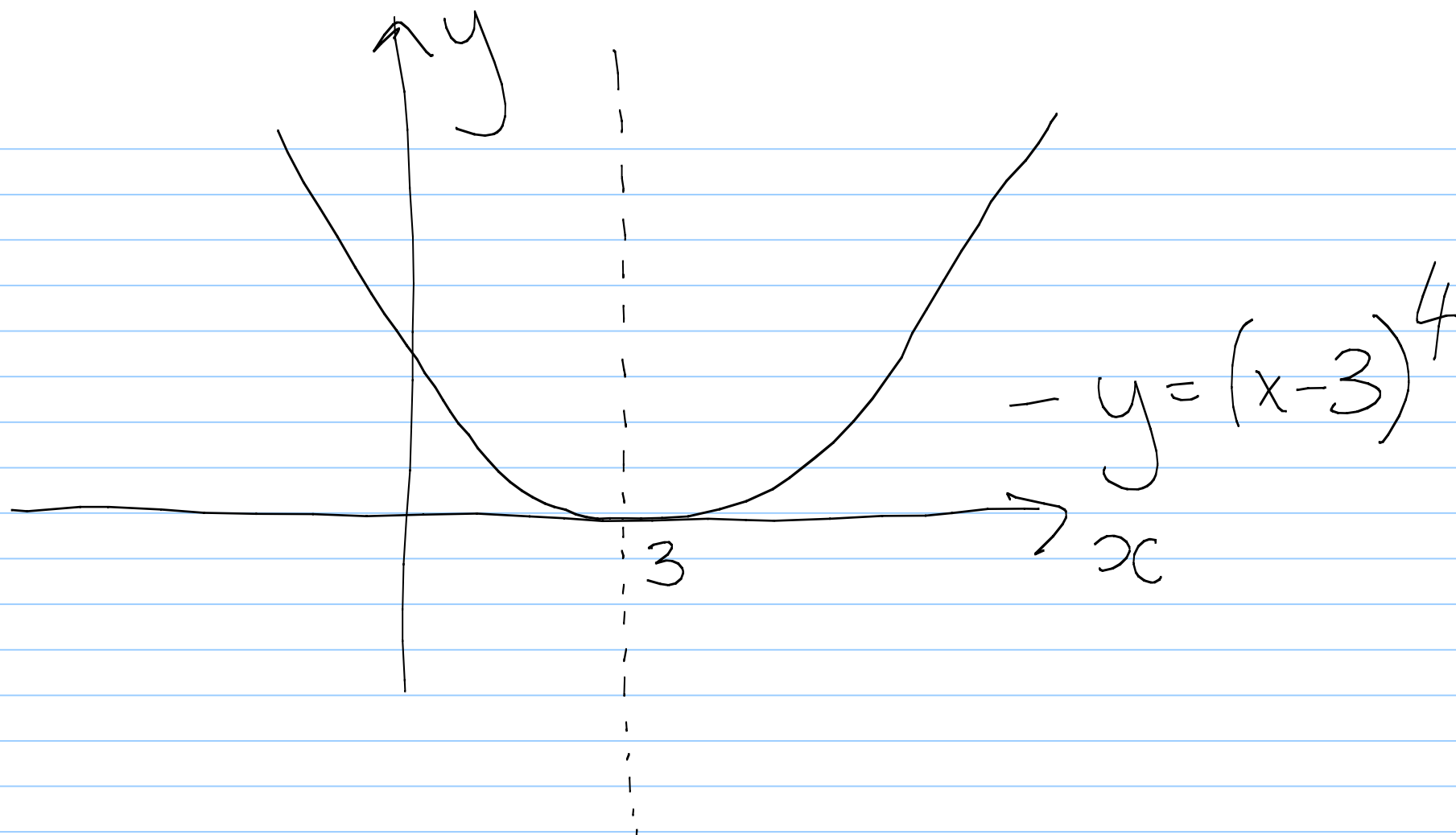
Example Sketch

NB: When you sketch shifted graphs, use a dotted line to indicate the shift.

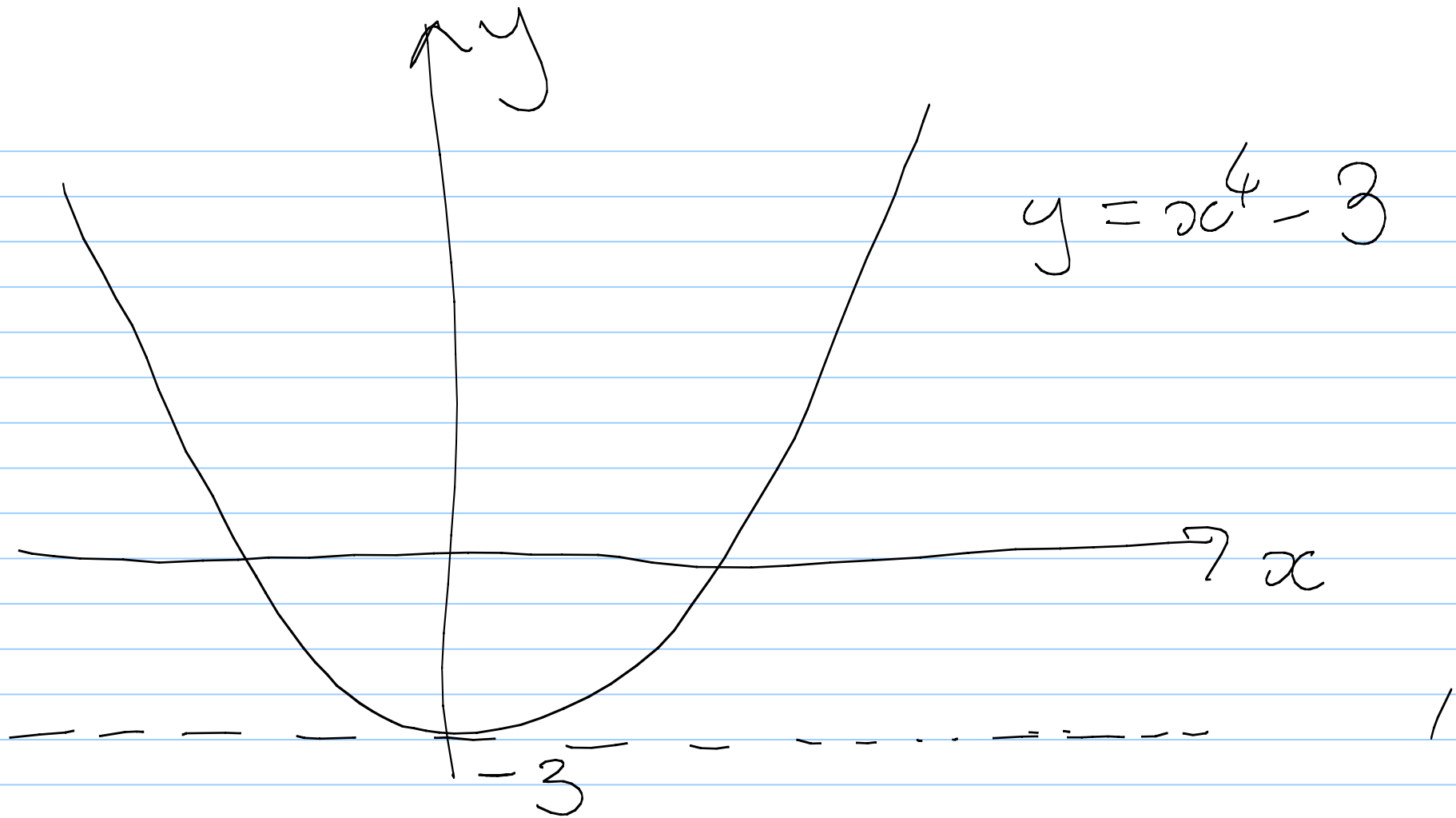
Example sketch

$$f(x) = (x-3)^4 \text{ and } g(x) = x^4 - 3$$



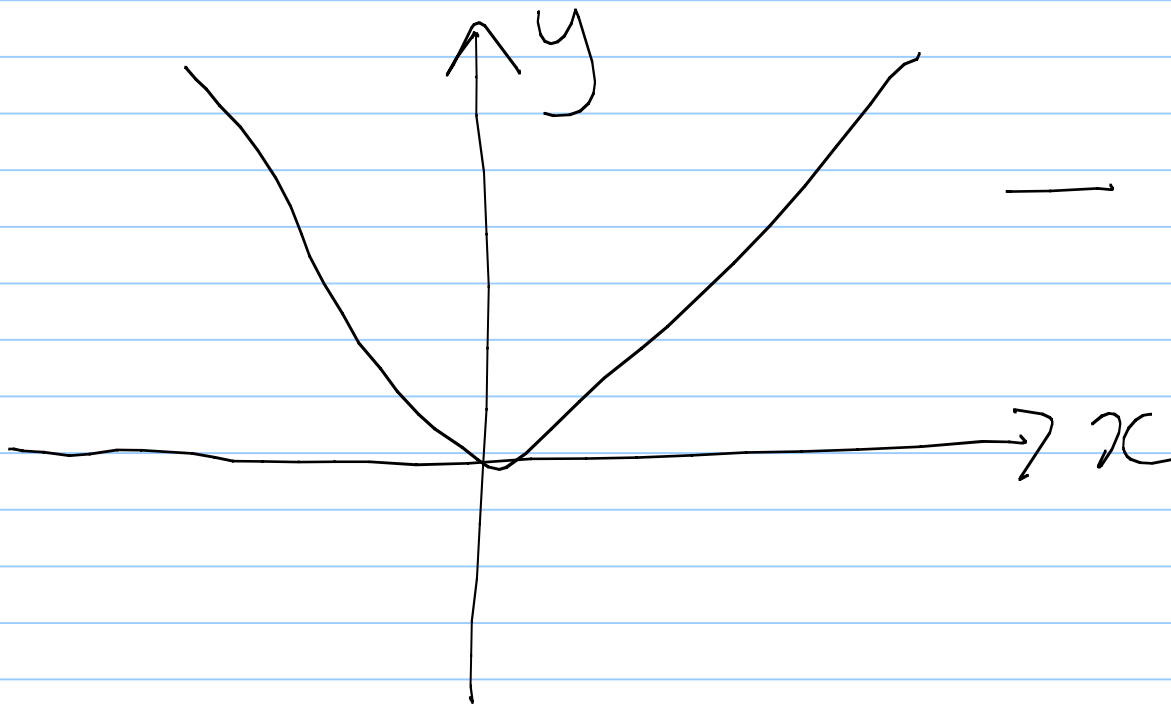


$$y = x^4 - 3$$

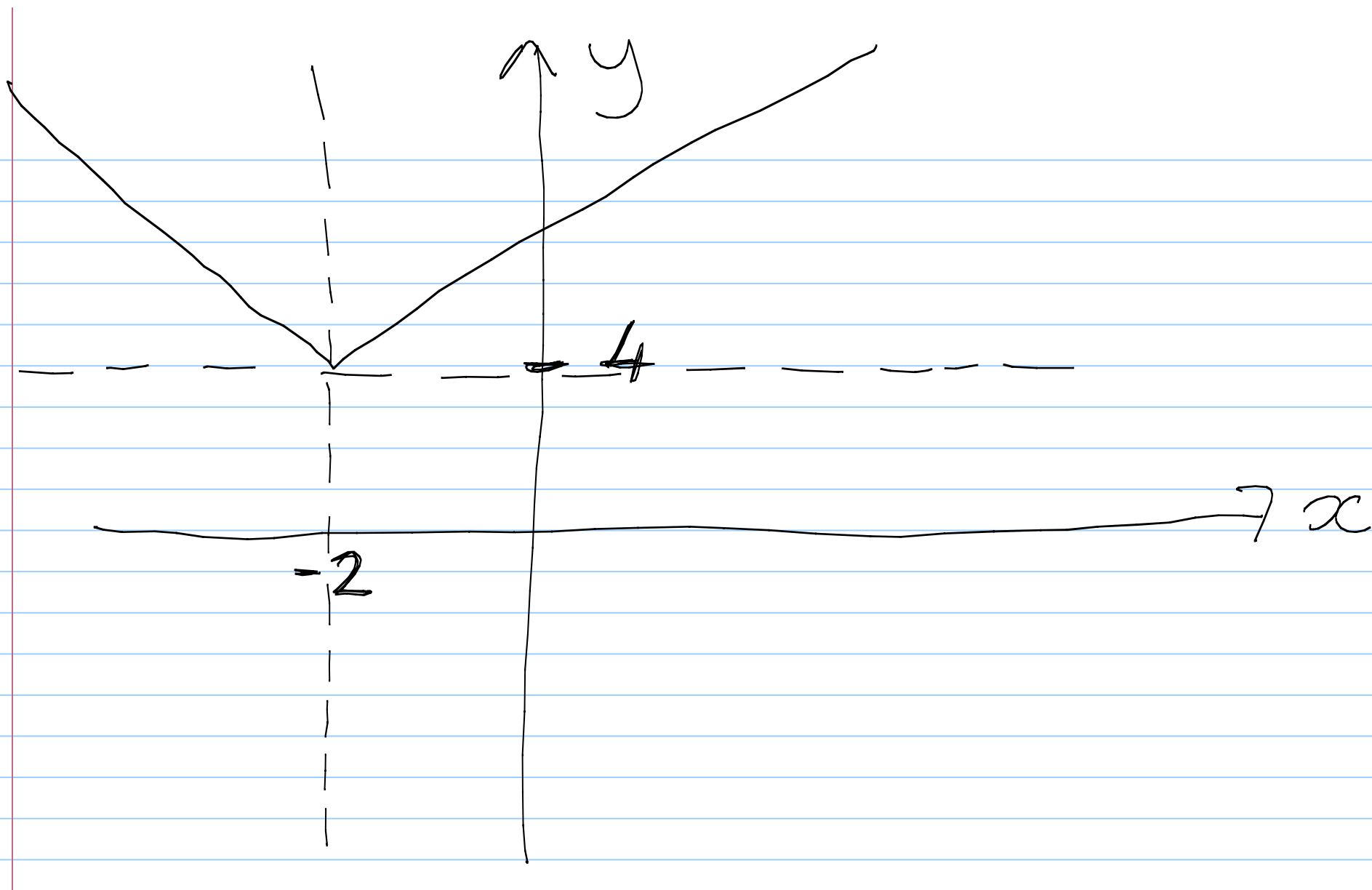


Example

Sketch $f(x) = |x+2| + 4$

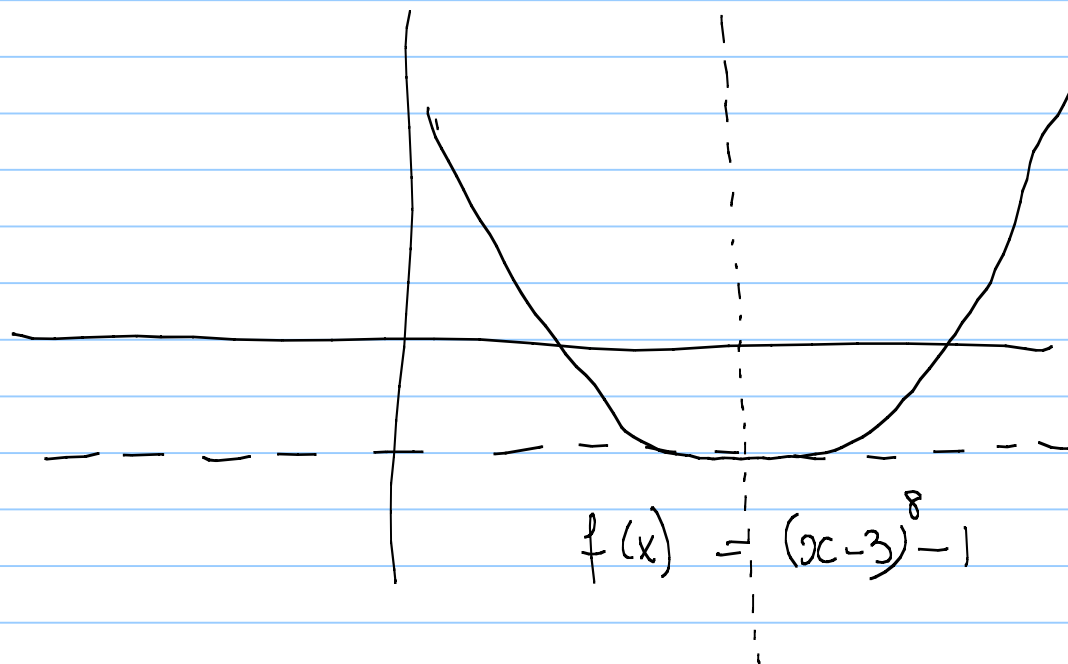
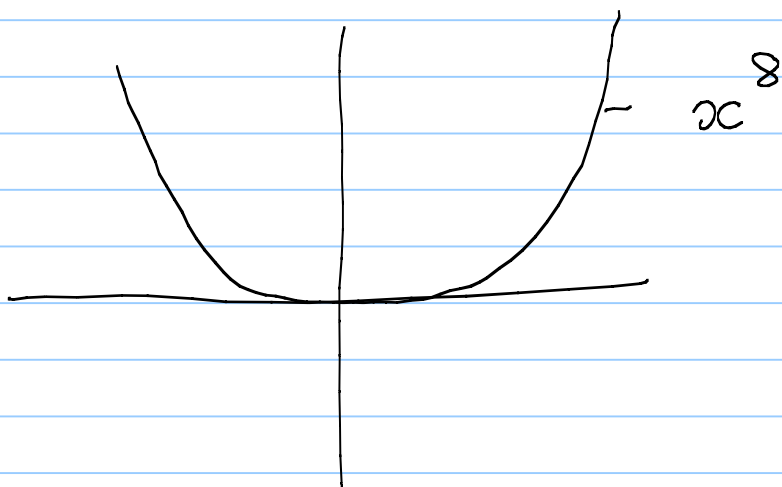


$$y = |x|$$



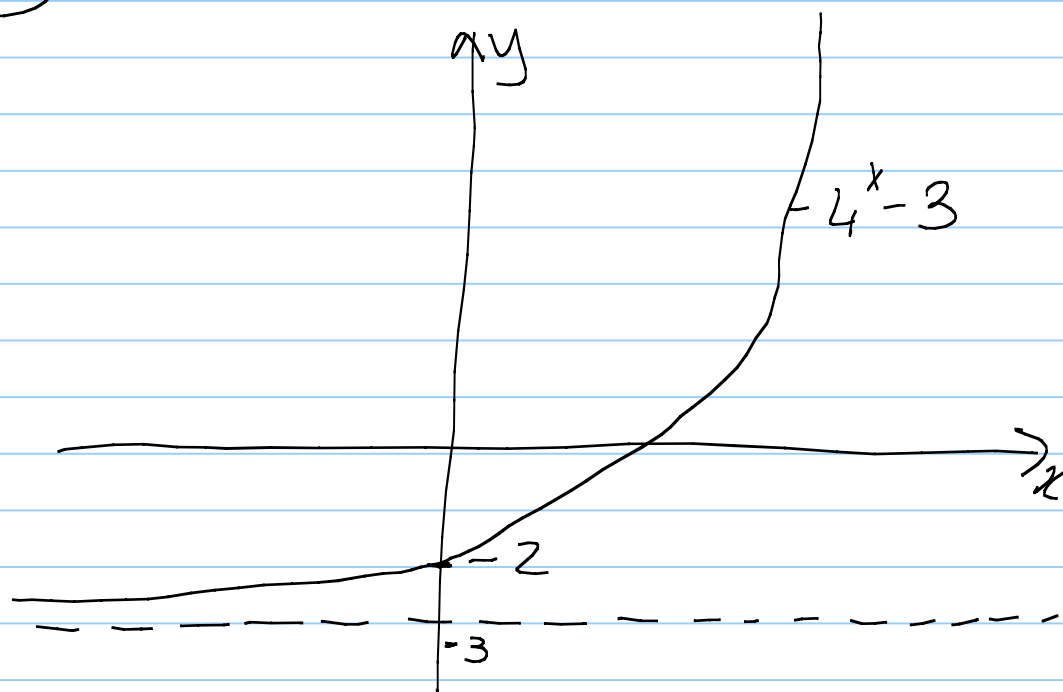
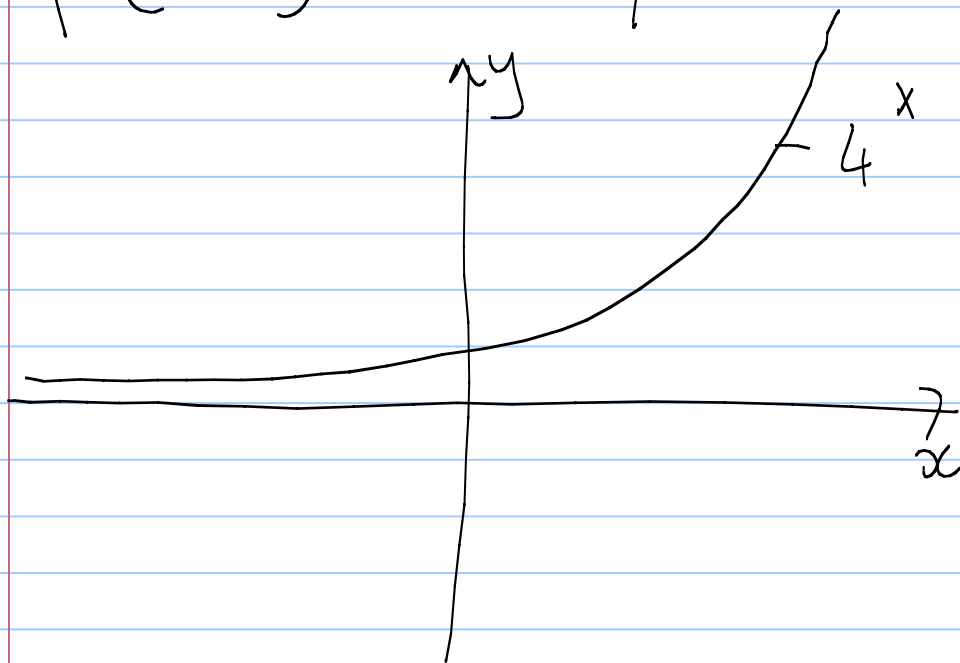
Example

① $f(x) = (x-3)^8 - 1$



Example

$$f(x) = 4^x - 3$$



Vertical and Horizontal Stretching
& reflecting

Suppose $c > 1$. To obtain the graph
of

* $y = c f(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c .

* $y = \left(\frac{1}{c}\right) f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of c .

* $y = f(cx)$, shrink the graph of $y = f(x)$ horizontally by a factor of c .

* $y = f\left(\frac{x}{c}\right)$, stretch the graph of $y = f(x)$ horizontally by a factor of c .

* $y = -f(x)$, reflect the graph of $y = f(x)$ about the x -axis.

* $y = f(-x)$, reflect the graph about the y -axis.

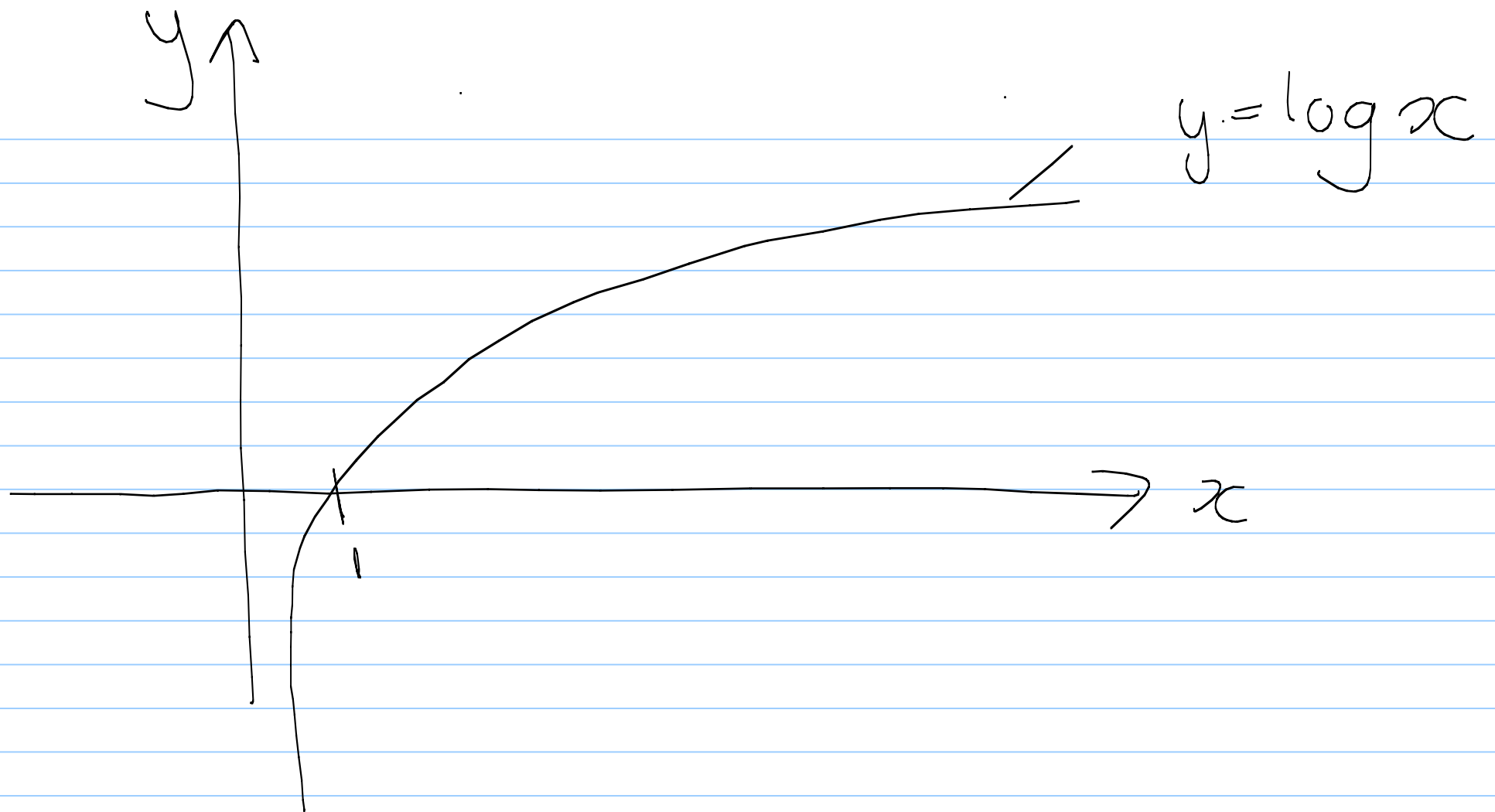
Example

Reflections

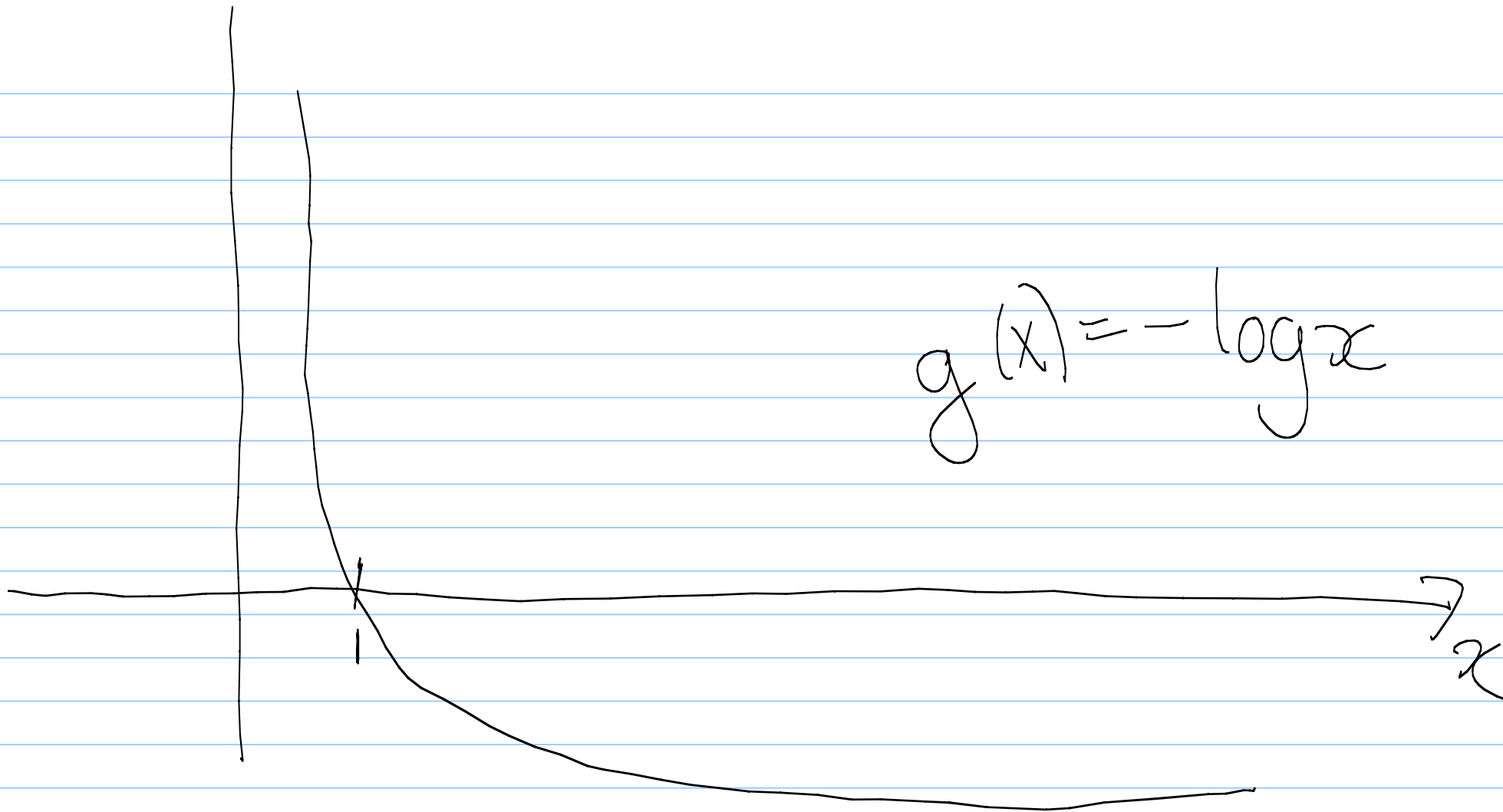
Sketch

$$f(x) = \log x, \quad g(x) = -\log x \quad \text{and}$$

$$h(x) = \log(-x)$$



$$g(x) = -\log x$$

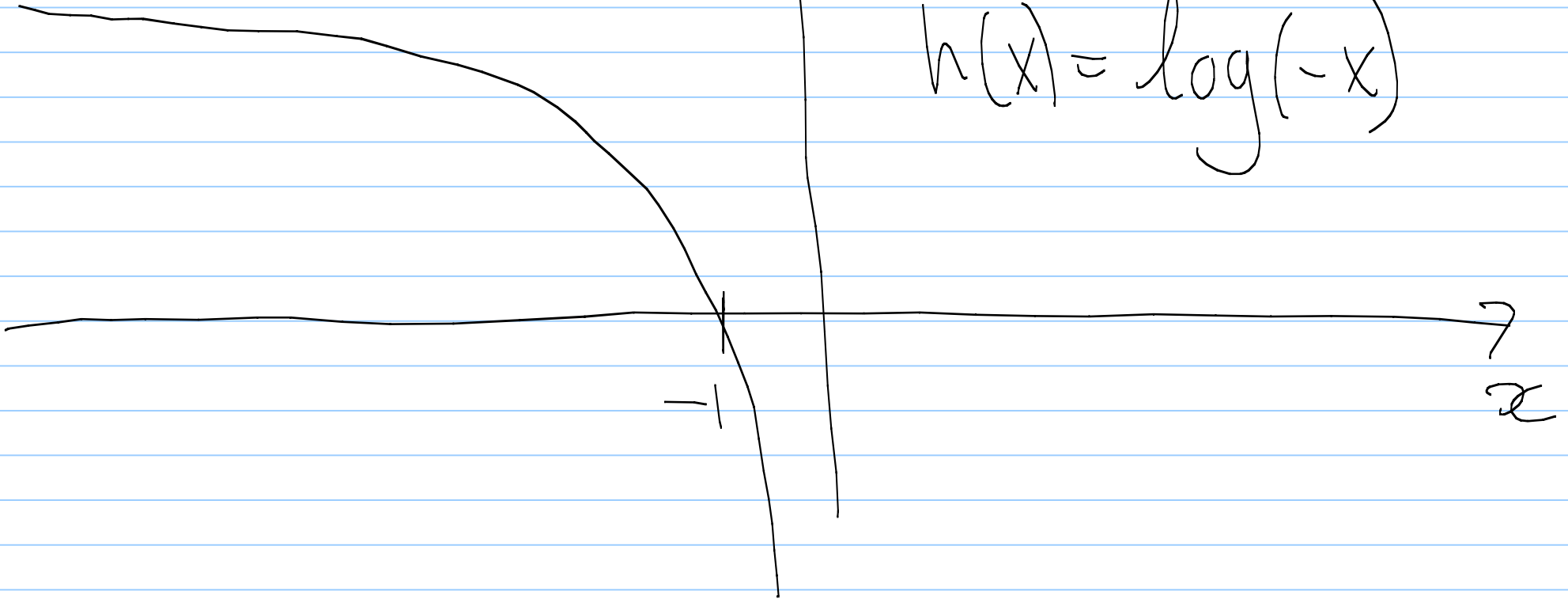


y

$$h(x) = \log(-x)$$

x

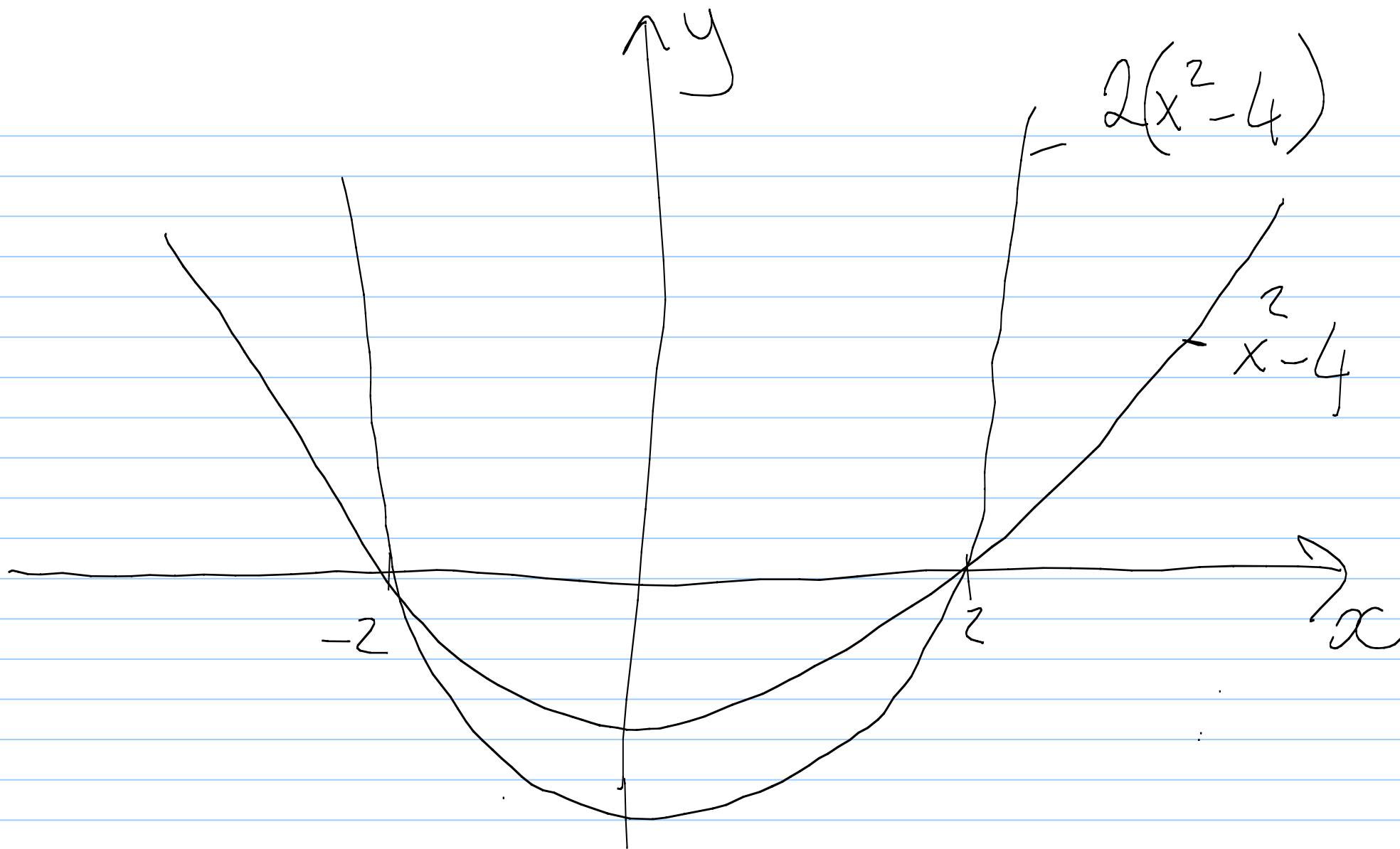
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Stretching and compressing

Sketch the graphs

$$y = x^2 - 4 \quad \text{and} \quad f(x) = 2(x^2 - 4)$$



When working with $\sin x$ & $\cos x$ functions.

$$y = a \sin(Bx)$$

$$y = a \cos(Bx)$$

we call

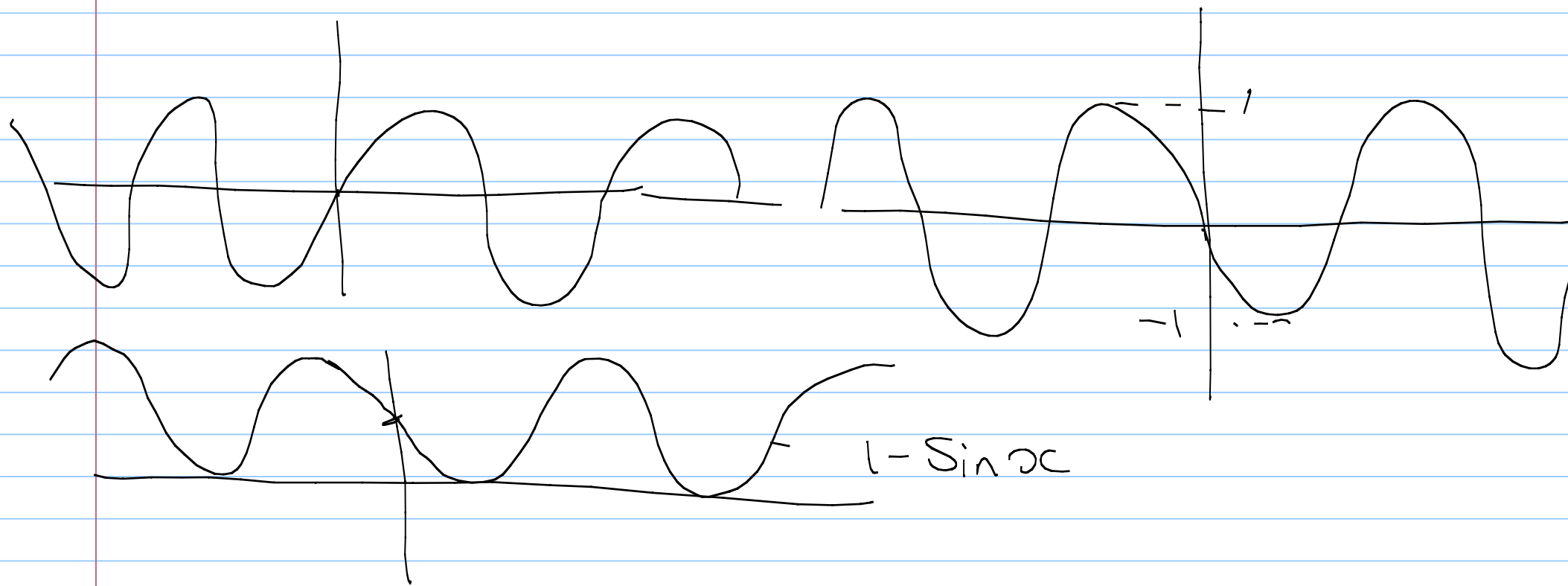
a : amplitude. & $\frac{2\pi}{|B|}$: period

Sketch

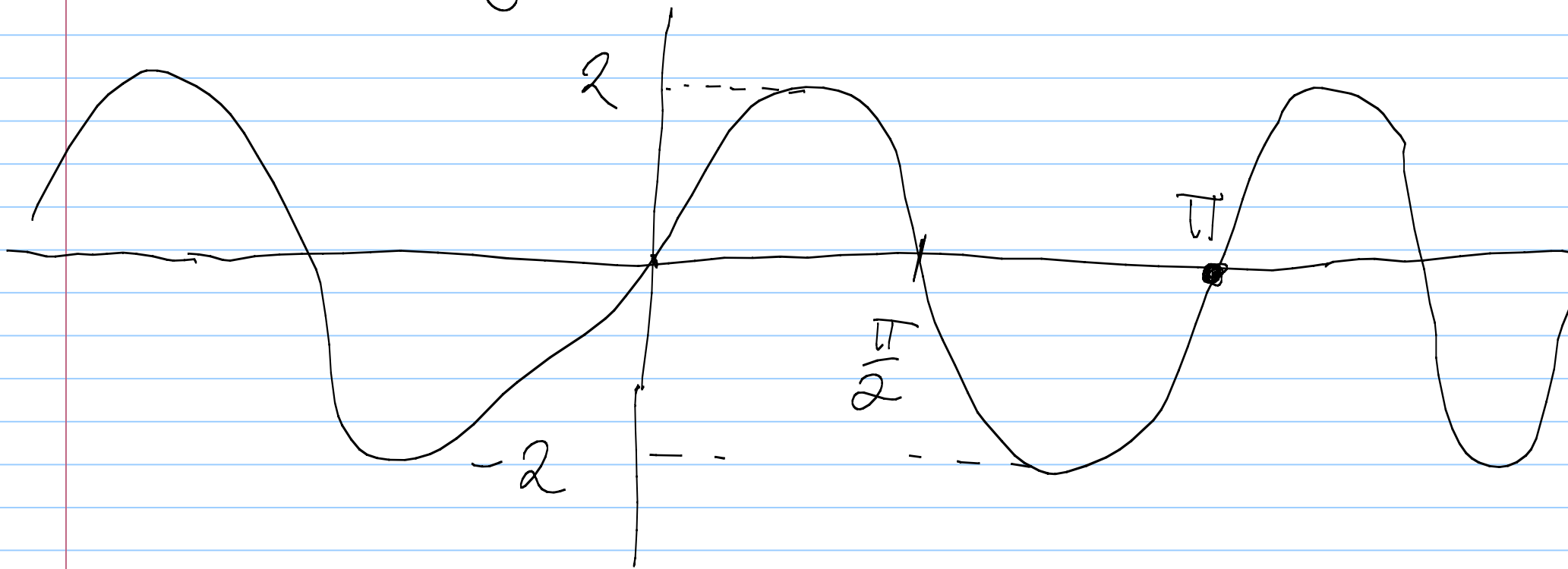
$$y = 1 - \sin x$$

$\sin x$

$-\sin x$



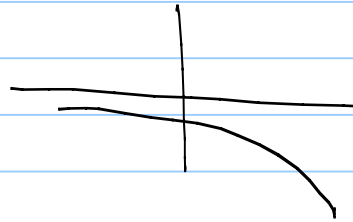
Example $y = 2 \sin 2x$ sketch.



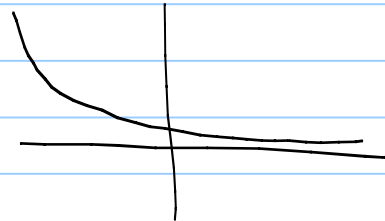
Examples

Sketch.

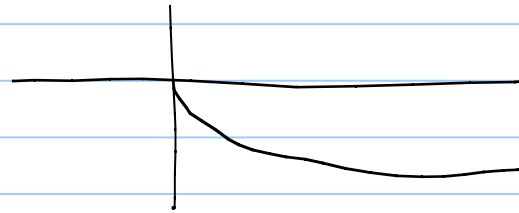
① $y = -2^x$



② $y = 2^{-x}$

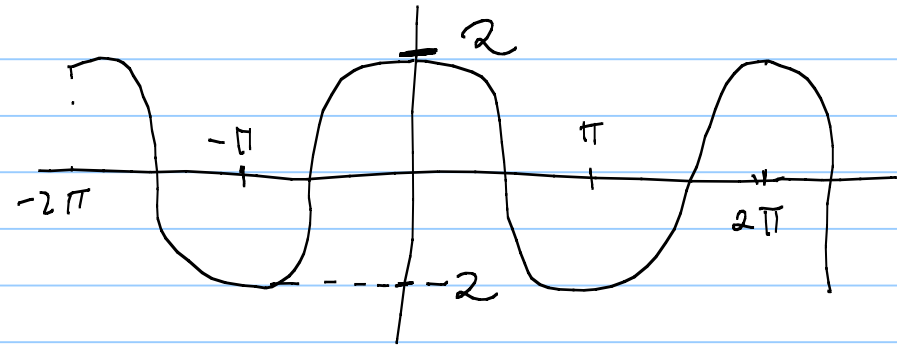


③ $y = -\sqrt{x}$



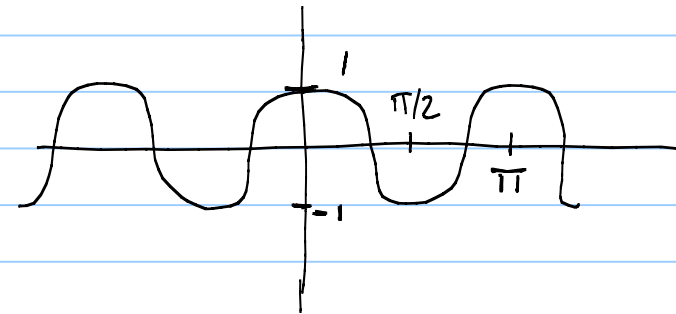
④

$$y = 2 \cos x$$



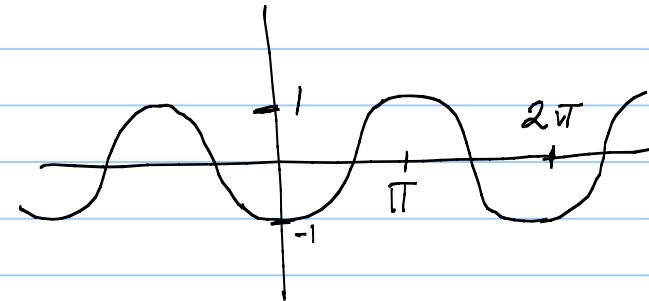
⑤

$$y = \cos(2x)$$



⑥

$$y = -\cos x$$



Combinations of Functions

* Two functions f & g can be combined to form new functions.

$$f+g, f-g, fg, f/g, /g \neq 0$$

Example.

$$f(x) = x \quad \& \quad g(x) = x^2$$

$$(f+g)(x) = x + x^2$$

$\longrightarrow$ Rest self study.

Composite functions p40 NB

Def: Given two functions f & g , the composite function $f \circ g$ is defined by

$$f \circ g(x) = f(g(x))$$

Example If $f(x) = x+1$ and $g(x) = \cos x$

Then $f \circ g(x) = \cos x + 1$

and

$$g \circ f(x) = \cos(x+1)$$

Example $f(x) = \sqrt{x}$ & $g(x) = \sqrt{2-x}$

Find $f \circ g(x)$ & $g \circ f(x)$.

$$f \circ g(x) = \sqrt{\sqrt{2-x}}$$

$$g \circ f(x) = \sqrt{2 - \sqrt{x}}$$

Example If $f(x) = x^2$ & $g(x) = \sin x$

Find $f \circ g(x)$ & $g \circ f(x)$.

$$f \circ g(x) = (\sin x)^2 = \sin^2 x$$

$$g \circ f(x) = \sin x^2$$

The functions are not the same.

Example If $g \circ f(x) = \sqrt[3]{\sqrt{x} + 1}$.

\$ f(x) = \sqrt[3]{x}\$ find $g(x)$

Then

$$g(x) = \sqrt[3]{x + 1}$$

• If $g \circ f(x) = 2^{x^2}$ & $g(x) = 2^x$
Find $f(x)$

Then $f(x) = x^2$

Example Find functions f and g such that

$$F = f \circ g$$

① $\frac{1}{x-3}$

③ $\sqrt{\tan x}$

② $\cos \sqrt{x}$

Solutions:

$$\textcircled{1} \quad f(x) = \frac{1}{x}$$

$$\text{and } g(x) = x - 3$$

$$\textcircled{2} \quad f(x) = \cos x \quad \text{and } g(x) = \sqrt{x}$$

$$\textcircled{3} \quad f(x) = \sqrt{x} \quad \text{and } g(x) = \tan x$$

Composite functions of 3 functions

$$f \circ g \circ h(x) = f(g(h(x)))$$

Example

$$f(x) = \frac{x}{x+1}$$

$$g(x) = \sin x$$

$$h(x) = x^2 + 1 \quad \text{Find } f \circ g \circ h(x)$$

$$f \circ g \circ h(x) = \frac{\sin(x^2+1)}{\sin(x^2+1) + 1}$$

Example

$$\text{If } f \circ g \circ h(x) = \cos^2(\sqrt{x})$$

find $f(x)$, $g(x)$ & $h(x)$

$$f \circ g \circ h(x) = [\cos \sqrt{x}]^2$$

$$f(x) = x^2$$

$$g(x) = \cos x$$

$$h(x) = \sqrt{x}$$

Example Write down

functions f, g, h such that

$$F = f \circ g \circ h \quad \text{if} \quad F(x) = \sqrt{2^{3-x}}$$

Then

$$f(x) = \sqrt{x}$$

$$g(x) = 2^x$$

$$h(x) = 3 - x$$